

F-TALENT CODE 2026

Segment Tree Hotel Queries

Time limit: 1.00 s | Memory limit: 512 MB



Problem Statement

There are n hotels lined up along a single street, numbered from 1 to n in the order they appear. For every hotel you are given the exact number of currently free rooms. Your job is to accommodate successive groups of tourists. Each group insists that all of its members must stay together in the same hotel; splitting a group across multiple hotels is not allowed.

The groups arrive one after another. For every group you are told in advance how many rooms it needs. When a group arrives you must assign it to the leftmost hotel (the one with the smallest index) that still has at least as many free rooms as the group requires. Once the assignment is made, the number of free rooms in that hotel is immediately reduced by the size of the group. If no hotel has enough free rooms for the current group, the group cannot be accommodated and you simply record a failure for that group.

Because the number of hotels and the number of groups can both be as large as two hundred thousand, a naïve linear scan for every group would be far too slow. An efficient solution therefore maintains a segment tree over the array of free-room counts. Each node of the tree stores the maximum number of free rooms present in its corresponding segment. With this structure it is possible, in logarithmic time, to locate the leftmost hotel whose free-room count is at least the required number, and then to decrease that count by a point update. The overall algorithm thus processes all groups in $O((n + m) \log n)$ time, which comfortably meets the tight time limits while correctly producing, for every group, either the 1-based index of the chosen hotel or zero if no suitable hotel exists.

Input

The first input line contains two integers n and m : the number of hotels and the number of groups.

The hotels are numbered 1, 2, ..., n

The next line contains n integers $h_1, h_2, \dots, h_n$: the number of free rooms in each hotel.

The last line contains m integers $r_1, r_2, \dots, r_m$: the number of rooms each group requires.

Output

Print the assigned hotel for each group. If a group cannot be assigned a hotel, print 0 instead.

Constraints

$$1 \leq n, m \leq 2 \cdot 10^5$$

$$1 \leq h_i \leq 10^9$$

$$1 \leq r_i \leq 10^9$$

Example 1:

INPUT	OUTPUT
8 5 3 2 4 1 5 5 2 6 4 4 7 1 1	3 5 0 1 1

Example 2:

INPUT	OUTPUT
10 10 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2	0 0 0 0 0 0 0 0 0 0

Example 3:

INPUT	OUTPUT
10 10 1 1 1 1 1 1 1 1 1 1000000000 2 2 2 2 2 2 2 2 2 2	10 10 10 10 10 10 10 10 10 10